The Chinese University of Hong Kong, SHenzhen

DDA 5002 Optimization

Sample Midterm Solutions

1. Solution:

(a). Let $v = |x_1 - c|$ and $y = \max\{a_{11}x_1 + a_{12}x_2, a_{21}x_1 - a_{22}x_2\}$. The original constraint is equivalent to $v + y \le b$. We can rewrite this single non-linear constraint as a set of linear constraints by introducing the auxiliary variables v and y.

$$\begin{aligned} v + y &\le b \\ v &\ge x_1 - c \\ v &\ge -(x_1 - c) \\ y &\ge a_{11}x_1 + a_{12}x_2 \\ y &\ge a_{21}x_1 - a_{22}x_2 \\ x_1, x_2, v &\ge 0 \\ y &\text{ is free} \end{aligned}$$

(b). The non-linear program is:

$$\min_{a,b} \sum_{i=1}^{n} |y_i - (ax_i + b)|$$

We can transform this into a linear program using one of two methods.

Method 1: Introduce n auxiliary variables v_i, one for each absolute value term. Let $v_i = |y_i - (ax_i + b)|$ for $i = 1, \ldots, n$. The problem is equivalent to the following LP:

$$\begin{aligned} \min \quad & \sum_{i=1}^{n} v_i \\ \text{s.t.} \quad & v_i \ge y_i - (ax_i + b) && \forall i = 1, \ldots, n \\ & v_i \ge -(y_i - (ax_i + b)) && \forall i = 1, \ldots, n \\ & v_i \ge 0 && \forall i = 1, \ldots, n \end{aligned}$$

Method 2: Introduce $2n$ non-negative auxiliary variables, e_i^+ and e_i^-, for $i = 1, \ldots, n$. Let $e_i = y_i - (ax_i + b)$. We can represent e_i as the difference of its positive and negative parts: $e_i = e_i^+ - e_i^-$. With this definition, the absolute value is $|e_i| = e_i^+ + e_i^-$. The problem is equivalent to the following LP:

$$\begin{aligned} \min \quad & \sum_{i=1}^{n} (e_i^+ + e_i^-) \\ \text{s.t.} \quad & e_i^+ - e_i^- = y_i - ax_i - b && \forall i = 1, \ldots, n \\ & e_i^+, e_i^- \ge 0 && \forall i = 1, \ldots, n \\ & a, b \text{ unrestricted} \end{aligned}$$

$\square$

2. Solution:

Indices & Sets

$i \in I$ set of ingredients

$j \in J$ set of candy types

Parameters

b_i maximum amount of ingredient $i \in I$ that can be purchased [ounces]

k_i cost per ounce of ingredient $i \in I$ [$/ounce]

l_{ij} minimum amount of ingredient $i \in I$ in a bar of candy $j \in J$ [ounces/bar]

u_{ij} maximum amount of ingredient $i \in I$ in a bar of candy $j \in J$ [ounces/bar]

r_j revenue per bar of candy $j \in J$ [$/bar]

d_j minimum demand for candy $j \in J$ [bars]

D_j maximum demand for candy $j \in J$ [bars]

f_j required weight of a bar of candy $j \in J$ [ounces]

Decision Variables

x_j number of bars of candy type $j \in J$ to produce

y_{ij} total ounces of ingredient $i \in I$ used to produce all bars of candy type $j \in J$

Formulation

$$
\begin{array}{lll}
\max & \sum_{j \in J} r_j x_j - \sum_{i \in I} \sum_{j \in J} k_i y_{ij} & \\
\text{s.t.} & d_j \leq x_j \leq D_j & \forall j \in J \\
& \sum_{j \in J} y_{ij} \leq b_i & \forall i \in I \\
& \sum_{i \in I} y_{ij} = f_j x_j & \forall j \in J \\
& y_{ij} \geq l_{ij} x_j & \forall i \in I, \forall j \in J \\
& y_{ij} \leq u_{ij} x_j & \forall i \in I, \forall j \in J \\
& x_j \geq 0 & \forall j \in J \\
& y_{ij} \geq 0 & \forall i \in I, \forall j \in J
\end{array}
$$

□

3. Solution:

(a). We introduce one artificial variable a_1 to constraint (1b). The Phase-I LP formulation is:

$$\begin{aligned} \max \quad & -a_1 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 - x_4 + a_1 = 1 \\ & 2x_1 + 3x_2 + x_3 + x_5 = 8 \\ & 4x_1 + 4x_2 - 3x_3 + x_6 = 4 \\ & x_1, x_2, x_3, x_4, x_5, x_6, a_1 \geq 0. \end{aligned}$$

Starting basic feasible solution: Basic variables $x_B = \begin{bmatrix} a_1 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 4 \end{bmatrix}$. Non-basic variables are $\{x_1, x_2, x_3, x_4\}$. Initial basis matrix is $A_B = I$. Non-basic matrix is $A_N = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 2 & 3 & 1 & 0 \\ 4 & 4 & -3 & 0 \end{bmatrix}$.

The non-basic cost vector is $C_N = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$. The basic cost vector is $C_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$.

Reduced costs: $\bar{C}_N^\top = C_N^\top - C_B^\top A_B^{-1} A_N = [0, 0, 0, 0] - [-1, 0, 0] I A_N = [1, 1, 1, -1]$.

We select x_1 to enter the basis (since $\bar{c}_1 = 1 > 0$). The search direction is $d_B = -A_B^{-1} A_1 = -I \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ -4 \end{bmatrix}$.

Ratio test: $t = \min \left\{ \frac{1}{1}, \frac{8}{2}, \frac{4}{4} \right\} = 1$. The variable a_1 leaves the basis. The new basic solution is: $x_1 = 1$, $x_5 = 6$, $x_6 = 0$.

Phase-I terminates with objective value 0. The starting basic feasible solution for Phase-II is $x = (1, 0, 0, 0, 6, 0)$. The corresponding basis is $\{x_1, x_5, x_6\}$ and the basis matrix is $A_B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

(b). The basic variables are $\{x_1, x_3, x_4\}$. The non-basic variables are $\{x_2, x_5, x_6\}$, so $x_2 = 0, x_5 = 0, x_6 = 0$. This gives the basic solution $x_B = \begin{bmatrix} x_1 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 14/5 \\ 12/5 \\ 21/5 \end{bmatrix}$.

To check for optimality, we use the Phase-II objective $\max 4x_1 + 5x_2 + x_3$. Basic variables $x_B = (x_1, x_3, x_4)$, Non-basic $x_N = (x_2, x_5, x_6)$. $C_B = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$, $C_N = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$. $A_B =$

$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 4 & -3 & 0 \end{bmatrix}$, $A_N = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

Simplex multipliers $y^\top = C_B^\top A_B^{-1}$. We solve $y^\top A_B = C_B^\top$: $\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 4 & -3 & 0 \end{bmatrix} =$

$\begin{bmatrix} 4 & 1 & 0 \end{bmatrix} \implies y^\top = [0, 1.6, 0.2]$.

Reduced costs $\bar{C}_N^\top = C_N^\top - y^\top A_N = [5, 0, 0] - [0, 1.6, 0.2] \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} = [-0.6, -1.6, -0.2]$.

Since all reduced costs are ≤ 0, this solution is optimal.

(c). New objective: $\max 4x_1 + 6x_2 + x_3$. The basis B, basic solution x_B, and simplex multipliers $y^\top = [0, 1.6, 0.2]$ are unchanged, as $C_B = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$ is the same. The non-basic cost vector is now $C_N = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ (for $x_N = (x_2, x_5, x_6)$).

Recalculate reduced costs: $\bar{C}_N^\top = C_N^\top - y^\top A_N = [6, 0, 0] - [5.6, 1.6, 0.2] = [0.4, -1.6, -0.2]$.

Since $\bar{c}_2 = 0.4 > 0$, the current solution is not optimal.

The search direction d corresponds to x_2 entering the basis. For non-basic variables, $d_N = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. For basic variables, $d_B = -A_B^{-1} A_2$, where $A_2 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$. We solve $A_B d_B = -A_2$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 4 & -3 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -4 \end{bmatrix} \implies d_B = \begin{bmatrix} -1.3 \\ -0.4 \\ -0.7 \end{bmatrix}.$$

To find how far we can move, we perform the ratio test: $t = \min \left\{ \frac{x_i}{-d_i} \mid d_i < 0 \right\} = \min \left\{ \frac{2.8}{1.3}, \frac{2.4}{0.4}, \frac{4.2}{0.7} \right\} = \min \left\{ \frac{28}{13}, 6, 6 \right\} = \frac{28}{13}$. We can move $t = 28/13$, and x_1 will leave the basis.

$\square$

4. Solution:
Let $x_{ij} \in \{0,1\}$ be the decision variable, where $i \in \{1,2\}$ represents the plant type and $j \in \{a,b,c\}$ represents the region. $x_{ij} = 1$ if a plant of type i is built in region j, and 0 otherwise.

(a). At least one plant must be built in each region:

$$x_{1j} + x_{2j} \geq 1 \qquad \forall j \in \{a,b,c\}$$

(b). At least one plant of type 1 must be built:

$$x_{1a} + x_{1b} + x_{1c} \geq 1$$

(c). If region a contains a plant of type 1 ($x_{1a} = 1$) then region c must also contain a plant of type 1 ($x_{1c} = 1$). This is modeled as $x_{1a} \implies x_{1c}$:

$$x_{1c} \geq x_{1a}$$

(d). Region a cannot contain a plant of type 1 and of type 2. This means at most one can be built:

$$x_{1a} + x_{2a} \leq 1$$

(e). Region b can contain a plant of type 2 only if a plant of type 1 is built in region a and a plant of type 2 is built in region c. This is modeled as $x_{2b} \implies (x_{1a} \text{ AND } x_{2c})$:

$$2x_{2b} \leq x_{1a} + x_{2c}$$

(f). Region c contains a plant of type 2 if and only if region a and region b both contain a plant of type 2. This linearizes the nonlinear constraint $x_{2c} = x_{2a} \cdot x_{2b}$:

$$\begin{aligned} x_{2c} &\leq x_{2a} \\ x_{2c} &\leq x_{2b} \\ x_{2c} &\geq x_{2a} + x_{2b} - 1 \end{aligned}$$

(g). The objective is to maximize the number of regions that contain plants of both types. We introduce auxiliary binary variables $y_j \in \{0,1\}$ for $j \in \{a,b,c\}$, where $y_j = 1$ if region j contains both plant types, and 0 otherwise. This means $y_j = x_{1j} \cdot x_{2j}$. The formulation is:

$$\begin{aligned} \max \quad & \sum_{j \in \{a,b,c\}} y_j \\ \text{s.t.} \quad & 2y_j \leq x_{1j} + x_{2j} && \forall j \in \{a,b,c\} \\ & y_j \in \{0,1\} && \forall j \in \{a,b,c\} \end{aligned}$$

$\square$

5. Solution:

(a). Based on the problem description and table: $a_1 = 2$, $a_2 = 3$, $a_3 = 4$, $b = 6$. The primal LP is:

$$\begin{aligned} \max \quad & 2x_1 + 4x_2 + x_3 \\ \text{s.t.} \quad & x_1 + 3x_2 \leq 8 \\ & 2x_1 + x_2 \leq 6 \\ & x_2 + 4x_3 \leq 6 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

(b). The dual problem is:

$$\begin{aligned} \min \quad & 8\pi_1 + 6\pi_2 + 6\pi_3 && \text{(1a)} \\ \text{s.t.} \quad & \pi_1 + 2\pi_2 \geq 2 && \text{(1b)} \\ & 3\pi_1 + \pi_2 + \pi_3 \geq 4 && \text{(1c)} \\ & 4\pi_1 \geq 1 && \text{(1d)} \\ & \pi_1, \pi_2, \pi_3 \geq 0. && \text{(1e)} \end{aligned}$$

(c). The complementary slackness conditions are:

$$\begin{aligned} (x_1 + 2x_2 - 8)\pi_1 &= 0 & \qquad (\pi_1 + 2\pi_2 - 2)x_1 &= 0 \\ (2x_1 + x_2 - 6)\pi_2 &= 0 & \qquad (3\pi_1 + \pi_2 + \pi_3 - 4)x_2 &= 0 \\ (x_2 + 4x_3 - 6)\pi_3 &= 0 & \qquad (4\pi_3 - 1)x_3 &= 0 \end{aligned}$$

(d). We are given the optimal solution $x^* = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$. Since $x_1 = 2 > 0$, $x_2 = 2 > 0$, $x_3 = 1 > 0$, the dual constraints (from part c) must be tight:

$$\begin{aligned} \pi_1 + 2\pi_2 &= 2 \\ 3\pi_1 + \pi_2 + \pi_3 &= 4 \\ 4\pi_3 &= 1 \end{aligned}$$

Solving the system gives:

$$\begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \begin{bmatrix} 11/10 \\ 9/20 \\ 1/4 \end{bmatrix}$$

(e). Yes. The shadow price for raw material 1 is $\pi_1 = 1.1$. This is the marginal value of one additional kg of raw material 1. Since the profit (\$1.1) is greater than the cost (\$1), we should make the purchase.

(f). The basic variables are $x_B = \{x_1, x_2, x_3\}$. The basis matrix is:

$$A_B = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 4 \end{bmatrix}$$

The inverse basis matrix is:

$$A_B^{-1} = \begin{bmatrix} -0.2 & 0.6 & 0 \\ 0.4 & -0.2 & 0 \\ -0.1 & 0.05 & 0.25 \end{bmatrix}$$

We are increasing the RHS of the second constraint ($i = 2$). The change in x_B is $A_B^{-1}\Delta b_2$, where $\lambda e_2 = \Delta b_2$ We need to maintain $x_B \geq 0$. The new solution is $x'_B = x^*_B + \lambda A_B^{-1} e_2$.

$$\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0.6 \\ -0.2 \\ 0.05 \end{bmatrix} \geq 0$$

For $\lambda \geq 0$, we only need to check components where $(A_B^{-1}e_2)_j < 0$. This is $j = 2$.

$$2 + (-0.2)\lambda \geq 0 \implies 2 \geq 0.2\lambda \implies \lambda \leq \frac{2}{0.2} = 10$$

The increase λ can be at most 10. The largest value of b (which is b_2) is $b_{new} = 6 + 10 = 16$.

□